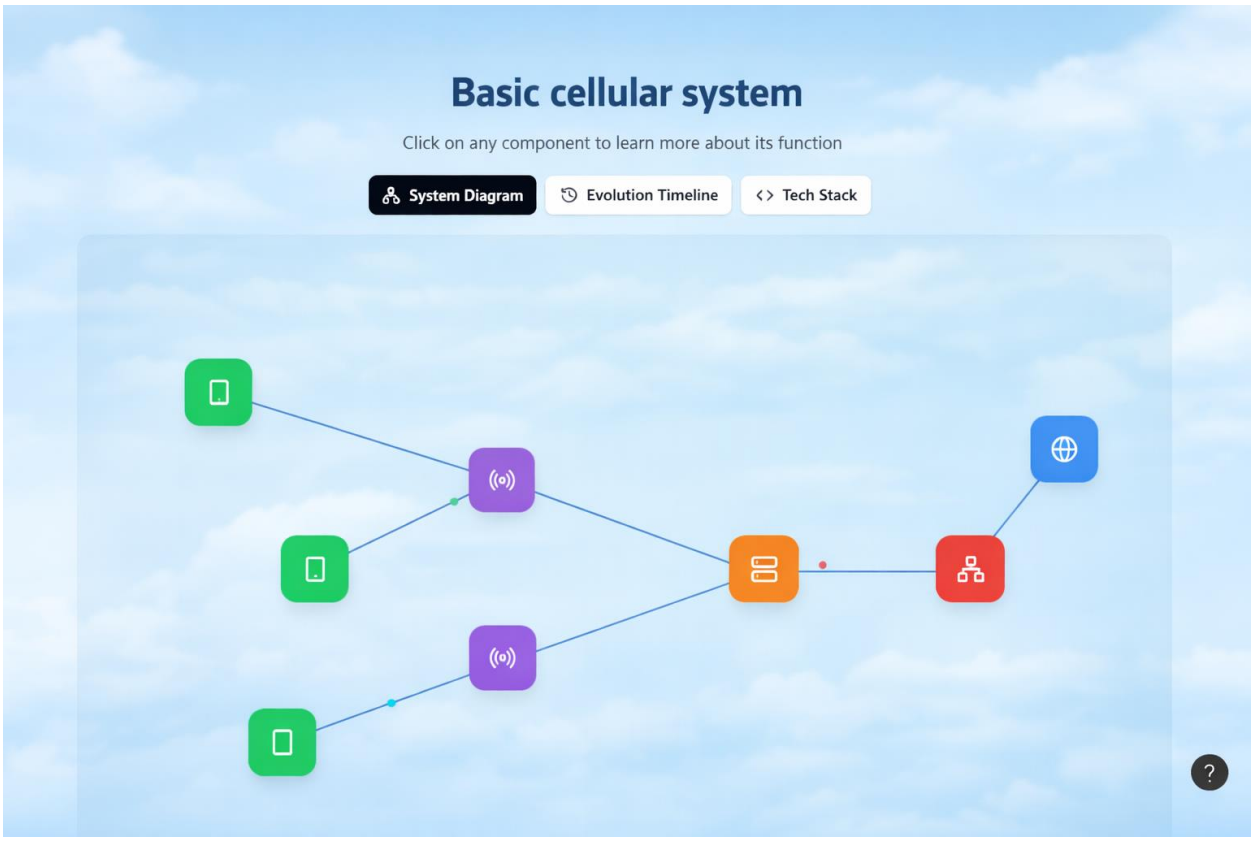
Basic cellular system
Click on any component to learn more about its function
System Diagram
Evolution Timeline
Tech Stack
?

Mobile Device (MS)

Mobile Station - The end-user device that communicates with the cellular network.

Key Functions:

- Transmits and receives radio signals
- Contains SIM card for subscriber identification
- Operates on assigned frequency bands
- Initiates and receives calls/data

Signal Flow: Mobile devices communicate via radio waves to the nearest BTS, which forwards the signal through the BSC to the MSC for routing.

Evolution Timeline
System Diagram
Evolution Timeline
Tech Stack
1G
1980s
2G
1991-2000s
3G
2001-2010s
4G LTE
2009-Present
4G LTE
2009-Present
5G
2019-Present
6G
(Future)
2G - 1991-2000s
Introduced digital transmission, enabling SMS, MMS, and basic data services.
Technology
Digital TDMA/CDMA
Max Speed
64 kbps (up to 384 kbps)
Key Features
Digital signals (GSM, CDMA)
SMS and MMS messaging
Basic data services (GPRS, EDGE)
Better voice quality and security
SIM card introduction

2G - 1991-2000s

Introduced digital transmission, enabling SMS, MMS, and basic data services.

Technology
Digital TDMA/CDMA

Max Speed
64 kbps (up to 384 kbps with EDGE)

Primary Use Case
Voice and text messajing

Key Features

- Digital signals (GSM, CDMA)
- SMS and MMS messaging
- Basic data services (GPRS, EDGE)
- Better voice quality and security
- SIM card introduction
- International roaming capability

Network Architecture

MS (Mobile Station) | BTS | BSC | MSC | HLR/VLR | PSTN/ISDN

Evolution Summary

Speed Evolution

- 1G: 2.4 kbps → 2G: 64 kbps
- 3G: 42 Mbps → 4G: 1 Gbps
- 5G: 10 Gbps → 6G: 1 Tbps

Key Transitions

- Analog → Digital (2G)
- Voice → Data (3G)
- Mobile → Everything (5G/6G)

Latency Improvement

- 3G: ~100ms
- 4G: ~50ms
- 5G: ~1ms